

Pedro G. S. Fernandes

Curriculum Vitae

 inspire-hep, Google Scholar, arXiv
 Personal Website
 GitHub

EMPLOYMENT

OCT 2026 –	FCT CEEC Postdoctoral Researcher CENTRA <i>Instituto Superior Técnico, Portugal</i>
OCT 2024 – SEP 2026	Postdoctoral Researcher Institut für Theoretische Physik <i>Heidelberg University, Germany</i>
OCT 2023 – SEP 2024	Postdoctoral Researcher Quantum Gravity Group <i>University of Southern Denmark, Denmark</i>
OCT 2022 – SEP 2023	Postdoctoral Researcher Particle Cosmology Group <i>University of Nottingham, UK</i>
NOV 2019 – JAN 2023	PhD Researcher LINK TO DISSERTATION Supervisors: David Mulryne and Timothy Clifton <i>Queen Mary University of London, UK</i>

EDUCATION

2017 – 2019	Master of Science in Physics LINK TO DISSERTATION Supervisors: Carlos Herdeiro and Eugen Radu <i>Instituto Superior Técnico, Portugal</i>
2014 – 2017	Bachelor of Physics <i>Instituto Superior Técnico, Portugal</i>

TEACHING EXPERIENCE

24-25 & 25-26	Quantum Field Theory <i>Teaching Assistant, MSc Course</i> <i>Heidelberg University, Germany</i>
2020-2023	Statistical Physics; Thermodynamics; Quantum Mechanics; Astrophysics <i>Teaching Assistant, BSc Course</i> <i>Queen Mary University of London, UK</i>
2019	Thermodynamics and Structure of Matter Excellence in teaching award <i>Teaching Assistant, BSc Course</i> <i>Instituto Superior Técnico, Portugal</i>

COMPUTER SKILLS AND CODE

C, C++, Python, Julia, Mathematica
Author of *SpinningBlackHoles*, a publicly available code to obtain stationary, axially-symmetric black hole solutions in modified gravity. See also my [GitHub](#).

SUMMARY OF PUBLICATIONS

Full list of publications available on my [inspire-hep](#) and [Google Scholar](#) profiles. **Bibliometric metrics:** 33 papers, 1753 citations, h-index=20 (data from inspire-hep). [Orcid](#): 0000-0002-8176-7208. [Scopus Author ID](#): 57209776012.

AWARDS AND FELLOWSHIPS

2026	FCT CEEC Fellowship <i>Instituto Superior Técnico</i>
OCT 2024	DFG Fellowship <i>Heidelberg University</i>
OCT 2023	Villum Fonden Fellowship <i>University of Southern Denmark</i>
OCT 2022	Leverhulme Trust Fellowship <i>University of Nottingham</i>
NOV 2019	Royal Society PhD Grant <i>Queen Mary University of London</i>
2019	FCT-CERN Research Grant <i>Collaborator, FCT-CERN</i>
2019	Excellence in Teaching Award <i>Instituto Superior Técnico</i>
FEB 2019 – JUL 2019	Teaching Fellowship <i>Instituto Superior Técnico</i>
JUL 2018 – JAN 2019	Research Fellowship SCIENTIFIC INITIATION GRANT Department of Physics <i>Instituto Superior Técnico</i>
JUN 2017 – JUN 2018	Research Fellowship SCIENTIFIC INITIATION GRANT Department of Physics <i>Instituto Superior Técnico</i>

ACADEMIC SERVICE

- Invited to referee for *Physical Review Letters*, *Physical Review D*, *Classical and Quantum Gravity*, *Physics Letters B*, *JCAP*, *European Physical Journal C*, *General Relativity and Gravitation*, and others.
- Member of the LISA consortium.
- Member of the Portuguese Society for Relativity and Gravitation.

STUDENT SUPERVISION AND MENTORING

- 2024-2025 Lidia Marino (MSc), co-advisor, now PhD student at Heidelberg University.
Sapienza University of Rome & Heidelberg University
Thesis title: “Different branches of scalarized black holes in a cubic scalar Gauss-Bonnet theory of gravity”
- 2024-2026 Fabian Willaschek (MSc), co-advisor, now PhD student at Heidelberg University
Heidelberg University
Thesis title: “Quantum fluctuations of gravity in Horndeski theory”
- 2025 Nadine Nussbaumer (PhD), informal co-advisor, now postdoc at the University of Gröningen
Heidelberg University
Thesis title: “Effective tales in gravity and cosmology”
- 2025-2026 Verena Köder (MSc), co-advisor, soon PhD student at Heidelberg University
Heidelberg University
Thesis title: “Hints of black-hole regularity in the EFT of gravity”

KNOWLEDGE TRANSFER

- Developer and maintainer of several open-source research codes, including [SpinningBlackHoles.jl](#), a comprehensive package for computing stationary black-hole solutions in alternative theories of gravity.
- Contributed to scientific outreach activities through *Circo da Física*, an initiative of the Núcleo de Física do Instituto Superior Técnico (NFIST), throughout my BSc and MSc studies.
- Contributed to public-facing articles communicating the results and broader significance of my research; see, for example, [this article](#) and [this feature](#).

PHD EXTERNAL EXAMINER

- 2026 Ulises Hernandez-Vera, University of Talca, Chile

EXTENDED RESEARCH VISITS

- JAN. 2026 Strong Gravity Group, Niels Bohr Institute, Denmark
- FEB. 2025 CENTRA, Instituto Superior Técnico, University of Lisbon, Portugal
- JUL. 2024 Perimeter Institute for Theoretical Physics, Canada
- FEB. 2024 IJCLab, Université Paris-Saclay, France

SELECTED CONFERENCES AND TALKS

- Crossing over the edge: workshop on regular black holes, SISSA, Trieste, Italy, invited speaker, Jul 2026
- Institut für Theoretische Physik, Frankfurt, invited seminar, Jun 2026
- SCALE: Strings and cosmology, Tenerife, Spain, speaker, May 2026
- Niels Bohr Institute, Denmark, invited seminar, Jan 2026
- XVIII Black Holes Workshop, Instituto Superior Técnico, Lisbon, speaker, Dec 2025
- Crossroads in Strong Gravity, University of Catania, Italy, session chair, Sep 2025
- University of Concepción, Chile, invited seminar, May 2025
- CENTRA, Instituto Superior Técnico, Lisbon, invited seminar, Feb 2025
- Gravity Shape Pisa 2024, invited keynote speaker, Oct 2024
- 50 Years of Horndeski Gravity, Perimeter Institute for Theoretical Physics, speaker, Jul 2024
- Université Paris-Saclay, IJCLab, invited seminar, Feb 2024
- AstroParticle Symposium 2023, Institut Pascal, Paris, invited keynote speaker, Nov 2023

- Nottingham Centre of Gravity Meeting, University of Nottingham, invited seminar, Apr 2023
- Imperial College London, invited seminar, Feb 2023
- XV Black Holes Workshop, ISCTE, Lisbon, keynote speaker, Dec 2022
- Gravity @ Prague, Charles University Prague, Participant, Sep 2022
- Center for Gravitation and Cosmology, Yangzhou University, invited webinar, Aug 2022
- London Cosmology Discussion Meeting (LCDM), invited speaker, Dec 2021
- 50th BUSSTEPP School, Queen Mary University of London, speaker, Jan 2021 (*award for best session talk*)
- XIII Black Holes Workshop, Instituto Superior Técnico, speaker, Dec 2020
- Quantum Gravity group, University of Groningen, invited webinar, May 2020
- Gravitational Geometry and Dynamics group, University of Aveiro, invited webinar, May 2020
- COSMONATA, Faculty of Sciences University of Lisbon, invited seminar, Dec 2019
- XI Black Holes Workshop, Instituto Superior Técnico, Lisbon, Participant, Dec 2018

LIST OF PUBLICATIONS

- [1] T. van der Steen, S. Maenaut, S. J. B. Husken, P. G. S. Fernandes, M. D. Jockwer, V. Cardoso, T. Hertog, and T. G. F. Li, Ringing of rapidly rotating black holes in effective field theory (2026), accepted for publication in Physical Review D., [arXiv:2604.11755 \[gr-qc\]](#) .
- [2] C. Charmousis, P. G. S. Fernandes, and M. Hassaine, Effective cosmological constant as black hole primary hair, *Phys. Rev. D* **113**, 124063 (2026), [arXiv:2603.25598 \[gr-qc\]](#) .
- [3] A. Eichhorn, P. G. S. Fernandes, and L. Marino, Strong breaking of black-hole uniqueness from coexisting scalarization mechanisms, *Phys. Rev. D* **113**, 124071 (2026), [arXiv:2603.05064 \[gr-qc\]](#) .
- [4] P. G. S. Fernandes, Exact analytic rotating black-hole solutions with primary hair (2026), submitted for publication in JCAP., [arXiv:2601.21163 \[gr-qc\]](#) .
- [5] P. G. S. Fernandes, Leading Effective Field Theory Corrections to the Kerr Metric at All Spins, *Phys. Rev. Lett.* **136**, 241401 (2026), [arXiv:2512.02338 \[gr-qc\]](#) .
- [6] P. G. S. Fernandes, J. Gou, L. Heisenberg, and N. Nussbaumer, Inflation, black holes with primary hair, and regular planar black holes from an infinite tower of regularized Lovelock-Proca corrections, *JCAP* **04**, 078, [arXiv:2511.22798 \[gr-qc\]](#) .
- [7] K. Destounis and P. G. S. Fernandes, Environmentally induced chaos: Extreme-mass-ratio systems of rotating black holes in astrophysical environments, *Phys. Rev. D* **113**, 044040 (2026), [arXiv:2508.20191 \[gr-qc\]](#) .
- [8] A. Eichhorn and P. G. S. Fernandes, Regular black holes without mass-inflation instability and gravastars from modified gravity, *Phys. Rev. D* **113**, L081501 (2026), [arXiv:2508.00686 \[gr-qc\]](#) .
- [9] P. G. S. Fernandes and V. Cardoso, Spinning Black Holes in Astrophysical Environments, *Phys. Rev. Lett.* **135**, 211403 (2025), [arXiv:2507.04389 \[gr-qc\]](#) .
- [10] F. Thaalba, P. G. S. Fernandes, and T. P. Sotiriou, Supermassive black hole scalarization and effective field theory, *Phys. Rev. D* **113**, 064005 (2026), [arXiv:2506.21434 \[gr-qc\]](#) .
- [11] R. Carballo-Rubio, H. Delaporte, A. Eichhorn, and P. G. S. Fernandes, Nonminimal light-curvature couplings and black-hole imaging, *Phys. Rev. D* **112**, 103016 (2025), [arXiv:2505.21431 \[astro-ph.HE\]](#) .
- [12] P. G. S. Fernandes and V. Cardoso, Dark matter as a vector field: an action principle for the Einstein cluster (2025), submitted for publication in Physical Review Letters., [arXiv:2505.00563 \[gr-qc\]](#) .
- [13] C. Charmousis, P. G. S. Fernandes, and M. Hassaine, Proca theory of four-dimensional regularized Gauss-Bonnet gravity and black holes with primary hair, *Phys. Rev. D* **111**, 124008 (2025), [arXiv:2504.13084 \[gr-qc\]](#) .
- [14] P. G. S. Fernandes, Regular BTZ black holes from an infinite tower of corrections, *Phys. Lett. B* **868**, 139772 (2025), [arXiv:2504.08565 \[gr-qc\]](#) .
- [15] P. G. S. Fernandes, Singularity resolution and inflation from an infinite tower of regularized curvature corrections, *Phys. Rev. D* **112**, 084028 (2025), [arXiv:2504.07692 \[gr-qc\]](#) .
- [16] K. V. Staykov, D. D. Doneva, P. G. S. Fernandes, and S. S. Yazadjiev, Rotating scalarized black holes: The role of the coupling, *Phys. Rev. D* **112**, 024058 (2025), [arXiv:2503.15206 \[gr-qc\]](#) .

- [17] A. Eichhorn and P. G. S. Fernandes, Purely metric Horndeski theories and spontaneous curvaturization of black holes, *Phys. Rev. D* **111**, 104007 (2025), [arXiv:2502.14717 \[gr-qc\]](#) .
- [18] G. Ventagli, P. G. S. Fernandes, A. Maselli, A. Padilla, and T. P. Sotiriou, Neutron stars and the cosmological constant problem, *Phys. Rev. D* **111**, 024001 (2025), [arXiv:2404.19012 \[gr-qc\]](#) .
- [19] P. G. S. Fernandes, C. Burrage, A. Eichhorn, and T. P. Sotiriou, Shadows and properties of spin-induced scalarized black holes with and without a Ricci coupling, *Phys. Rev. D* **109**, 104033 (2024), [arXiv:2403.14596 \[gr-qc\]](#) .
- [20] A. Eichhorn, P. G. S. Fernandes, A. Held, and H. O. Silva, Breaking black-hole uniqueness at supermassive scales, *Class. Quant. Grav.* **42**, 105006 (2025), [arXiv:2312.11430 \[gr-qc\]](#) .
- [21] C. Burrage, P. G. S. Fernandes, R. Brito, and V. Cardoso, Spinning black holes with axion hair, *Class. Quant. Grav.* **40**, 205021 (2023), [arXiv:2306.03662 \[gr-qc\]](#) .
- [22] P. G. S. Fernandes, Rotating black holes in semiclassical gravity, *Phys. Rev. D* **108**, L061502 (2023), [arXiv:2305.10382 \[gr-qc\]](#) .
- [23] P. G. S. Fernandes and D. J. Mulryne, A new approach and code for spinning black holes in modified gravity, *Class. Quant. Grav.* **40**, 165001 (2023), [arXiv:2212.07293 \[gr-qc\]](#) .
- [24] P. G. S. Fernandes, D. J. Mulryne, and J. F. M. Delgado, Exploring the Small Mass Limit of Stationary Black Holes in Theories with Gauss-Bonnet Terms, *Class. Quant. Grav.* **39**, 235015 (2022), [arXiv:2207.10692 \[gr-qc\]](#) .
- [25] P. G. S. Fernandes, P. Carrilho, T. Clifton, and D. J. Mulryne, The 4D Einstein–Gauss–Bonnet theory of gravity: a review, *Class. Quant. Grav.* **39**, 063001 (2022), [arXiv:2202.13908 \[gr-qc\]](#) .
- [26] P. G. S. Fernandes, P. Carrilho, T. Clifton, and D. J. Mulryne, Black holes in the scalar-tensor formulation of 4D Einstein-Gauss-Bonnet gravity: Uniqueness of solutions, and a new candidate for dark matter, *Phys. Rev. D* **104**, 044029 (2021), [arXiv:2107.00046 \[gr-qc\]](#) .
- [27] P. G. S. Fernandes, Gravity with a generalized conformal scalar field: theory and solutions, *Phys. Rev. D* **103**, 104065 (2021), [arXiv:2105.04687 \[gr-qc\]](#) .
- [28] T. Clifton, P. Carrilho, P. G. S. Fernandes, and D. J. Mulryne, Observational Constraints on the Regularized 4D Einstein-Gauss-Bonnet Theory of Gravity, *Phys. Rev. D* **102**, 084005 (2020), [arXiv:2006.15017 \[gr-qc\]](#) .
- [29] P. G. S. Fernandes, P. Carrilho, T. Clifton, and D. J. Mulryne, Derivation of Regularized Field Equations for the Einstein-Gauss-Bonnet Theory in Four Dimensions, *Phys. Rev. D* **102**, 024025 (2020), [arXiv:2004.08362 \[gr-qc\]](#) .
- [30] P. G. S. Fernandes, Charged black holes in AdS spaces in 4D Einstein Gauss-Bonnet gravity, *Phys. Lett. B* **805**, 135468 (2020), [arXiv:2003.05491 \[gr-qc\]](#) .
- [31] P. G. S. Fernandes, Einstein–Maxwell-scalar black holes with massive and self-interacting scalar hair, *Phys. Dark Univ.* **30**, 100716 (2020), [arXiv:2003.01045 \[gr-qc\]](#) .
- [32] P. G. S. Fernandes, C. A. R. Herdeiro, A. M. Pombo, E. Radu, and N. Sanchis-Gual, Charged black holes with axionic-type couplings: Classes of solutions and dynamical scalarization, *Phys. Rev. D* **100**, 084045 (2019), [arXiv:1908.00037 \[gr-qc\]](#) .
- [33] P. G. S. Fernandes, C. A. R. Herdeiro, A. M. Pombo, E. Radu, and N. Sanchis-Gual, Spontaneous Scalarisation of Charged Black Holes: Coupling Dependence and Dynamical Features, *Class. Quant. Grav.* **36**, 134002 (2019), [arXiv:1902.05079 \[gr-qc\]](#) .